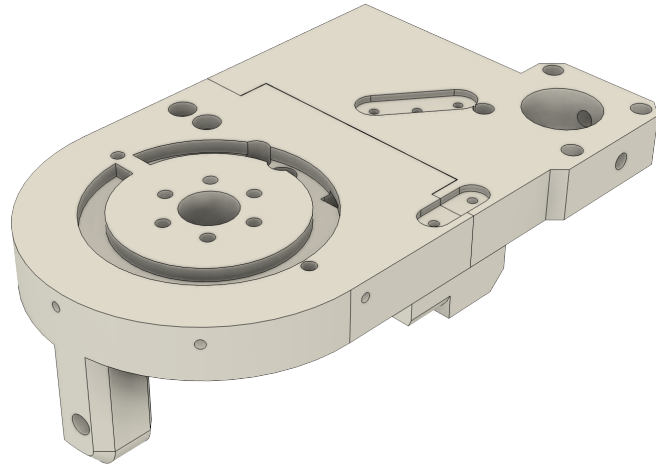
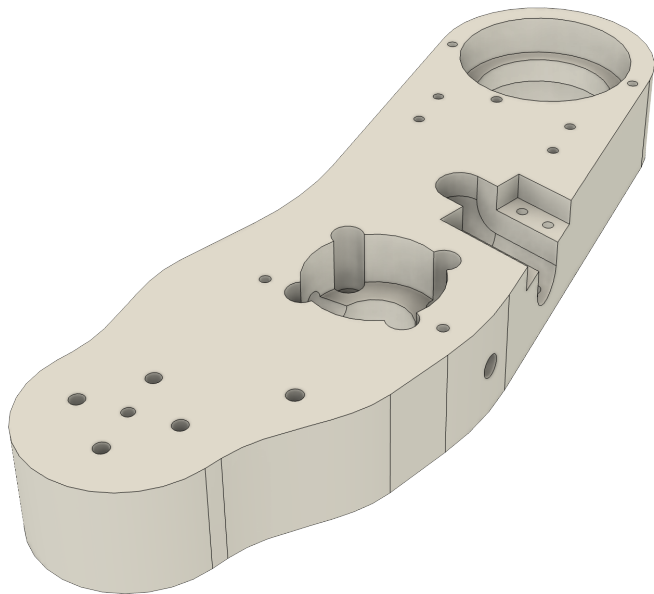


PAROL6 SMALL PRINT BED MODIFICATION

Splitting J1_turret_rework_blocker and Upper_arm
for printers with a bed smaller than 175 mm



About this modification



Parts affected

Upper_arm

102 x 231 x 28 mm

J1_turret_rework_blocker

114 x 175 x 75 mm

Both exceed a 150 mm cube.

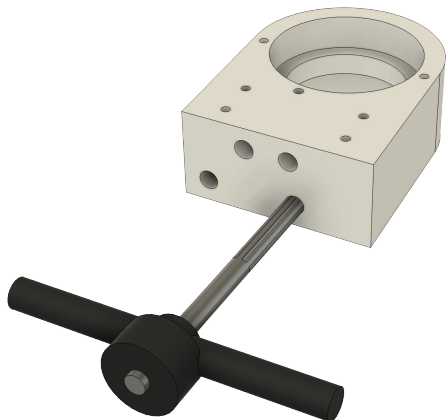
Two PAROL6 parts do not fit on a 150 mm printer. No rotation makes them fit: the smallest cube that can contain the upper arm is 160.5 mm, and the turret blocker 154.1 mm.

This modification splits each part in two, at a plane chosen to miss every screw hole. The halves are located with steel dowel pins and bonded with two part epoxy.

No other part of the robot changes. Assembly of the arm itself follows the main manual exactly.

Required materials

Tools required



A 4 mm and 6 mm drill bit turned by hand will work, but cuts oversize and less round.

Tools

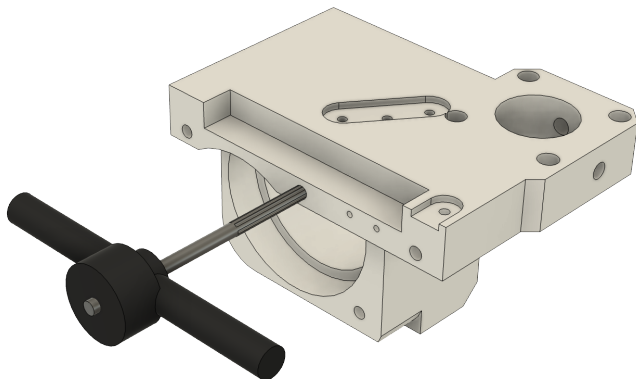
4 mm H7 chucking reamer
6 mm H7 chucking reamer
Tap wrench, 12 mm capacity
Hobby knife or deburring tool
Flat reference surface
4 to 6 spring clamps

Printed holes come out undersize and out of round, especially horizontal ones. Reaming gives the dowel pins a hole they can actually locate in.

Both reamers have round shanks, so you need a tap wrench with at least 12 mm capacity. Do not turn them in a power drill: PETG melts, the hole closes behind the cutter and the reamer grabs.

Turn clockwise going in AND clockwise coming out. Reversing a reamer dulls it and tears the hole.

Consumables



Consumables

3 x dowel pin 4 x 18 mm

4 x dowel pin 6 x 22 mm

Two part epoxy, slow set

120 grit sandpaper

99% isopropyl alcohol

Nitrile gloves, swabs, sticks

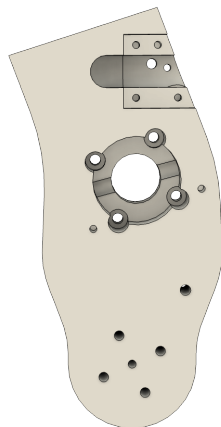
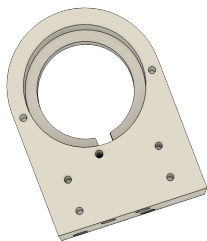
Pins must be SOLID dowel pins, stainless or hardened steel. Do not use roll pins or spring pins: they are rolled tube, they compress, and they cannot locate anything.

Use a slow setting two part epoxy. Five minute epoxy starts to gel before you have placed four pins and closed a 60 mm seam.

Do not use cyanoacrylate. It bonds PETG poorly and has no gap filling ability.

Printing the parts

Printing the parts



Printed parts

Upper_arm_A_long_end

137 x 136 mm

Upper_arm_B_short_end

82 x 63 mm

J1_turret_rework_blocker_A_turret_end

100 x 105 mm

J1_turret_rework_blocker_B_arm_end

114 x 80 mm

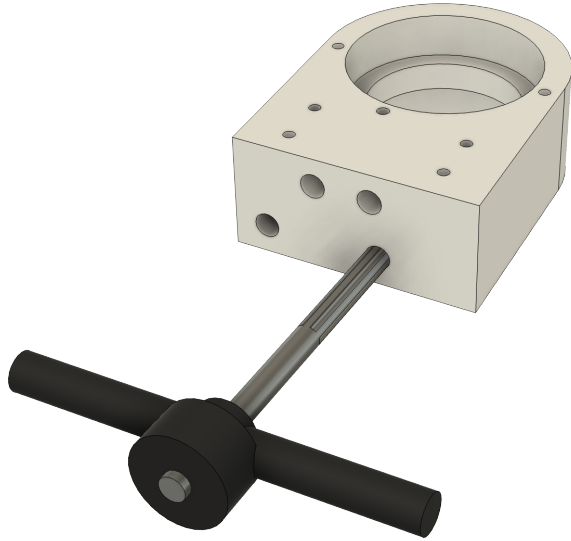
Print all four halves in PETG, using the infill and settings from the main printing table.

Lay each half with its largest flat face on the bed. Upper arm part A is the tightest at 137 x 136 mm, so rotate it on the bed and use a brim rather than a raft.

Clean off the brim and check for elephant foot before doing anything else. A raised first layer edge on a mating face will hold the seam open.

Upper arm

Step 1 - Upper arm - ream the dowel holes



This step

6 mm H7 reamer

Tap wrench

Part A 4 holes, 11 mm deep

Part B 4 holes, 13 mm deep

Total 8 holes

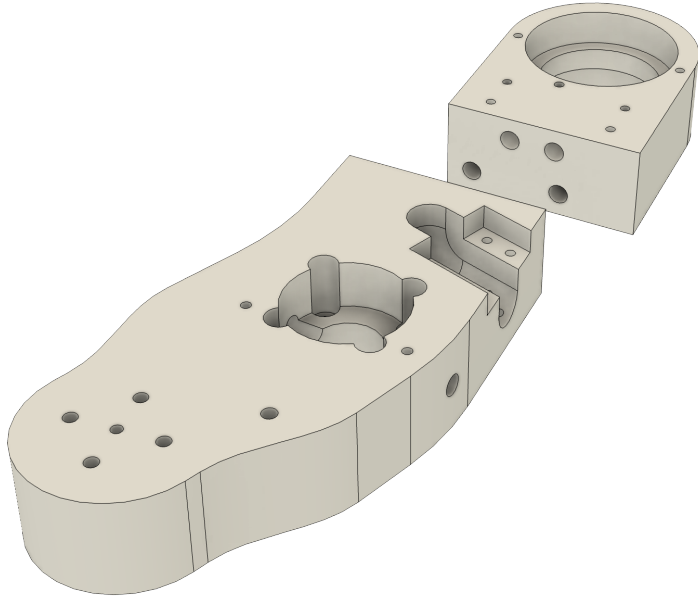
Ream all eight 6 mm holes, four in each half.

Hold the reamer square to the seam face and turn it steadily by hand. Back it out every few turns to clear chips, which come off PETG stringy and pack the flutes.

Clockwise in, clockwise out. Never reverse.

Test each hole with a pin. You want a firm slip fit: in under thumb pressure, staying put when you turn the part over.

Step 2 - Upper arm - sand and degrease



This step

120 grit sandpaper
99% isopropyl alcohol
Nitrile gloves

Sand both seam faces

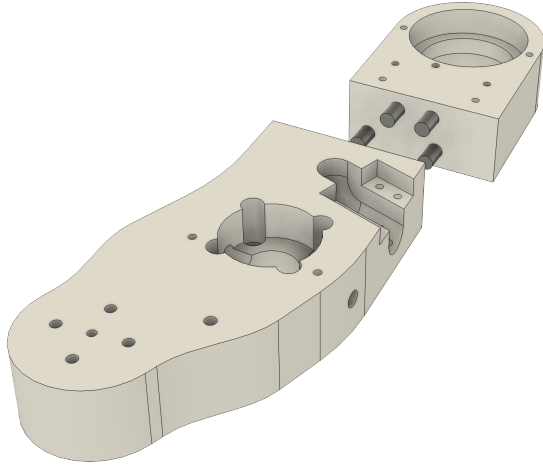
Scuff both seam faces with 120 grit until they are uniformly dull. Any patch left glossy is a patch the epoxy will not grip.

This step does more for joint strength than the choice of adhesive.

Do not sand inside the pin holes, you just reamed them.

Wipe both faces with isopropyl, let it flash off, then keep bare fingers off them. Gloves on from here.

Step 3 - Upper arm - dry fit



If the seam will not close, find out why now. Usually elephant foot on a mating face.

This step

Upper_arm_A_long_end
Upper_arm_B_short_end
4 x dowel pin 6 x 22 mm

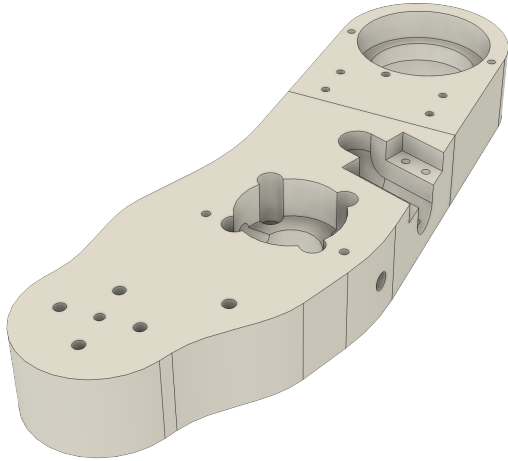
No glue at this stage

Seat all four pins in part B first. B takes 13 mm of a 22 mm pin, leaving 9 mm to enter an 11 mm hole in A, so the pins can never bottom out and hold the seam open.

Push the halves together by hand. Look along the seam from several angles: it should close with no visible gap and no light through it.

Stand the assembly on a flat surface and check it does not rock.

Step 4 - Upper arm - glue and clamp



Wipe squeeze out with isopropyl NOW. Cured epoxy is very hard to remove from PETG.

This step

Two part epoxy
4 x dowel pin 6 x 22 mm
Spring clamps or tape

Set 4-6 h, full cure 24 h

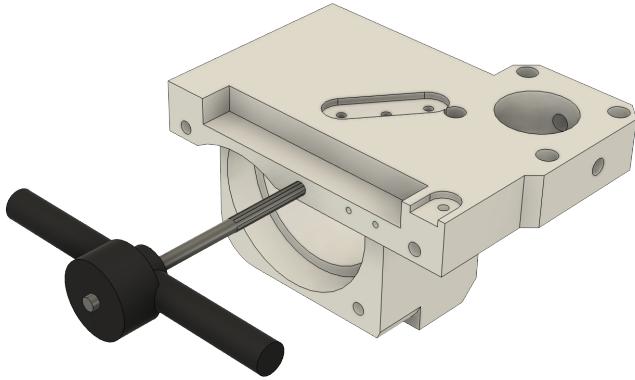
Mix the epoxy for a full two minutes until the colour is even with no streaks. You need well under 1 ml.

Thin coat on BOTH seam faces. Wipe a light film on each pin and drop it in. Do NOT fill the pin holes: a close pin in a blind hole full of paste epoxy hydraulic locks and stops short.

Close the joint and clamp perpendicular to the seam face, not along the part. Tape across the joint handles the 18 degree angle better than a clamp. Snug, not crushing.

J1 turret

Step 1 - J1 turret - ream the dowel holes



This step

4 mm H7 reamer

Tap wrench

Part A 3 holes, 7 mm deep

Part B 3 holes, 12 mm deep

Total 6 holes

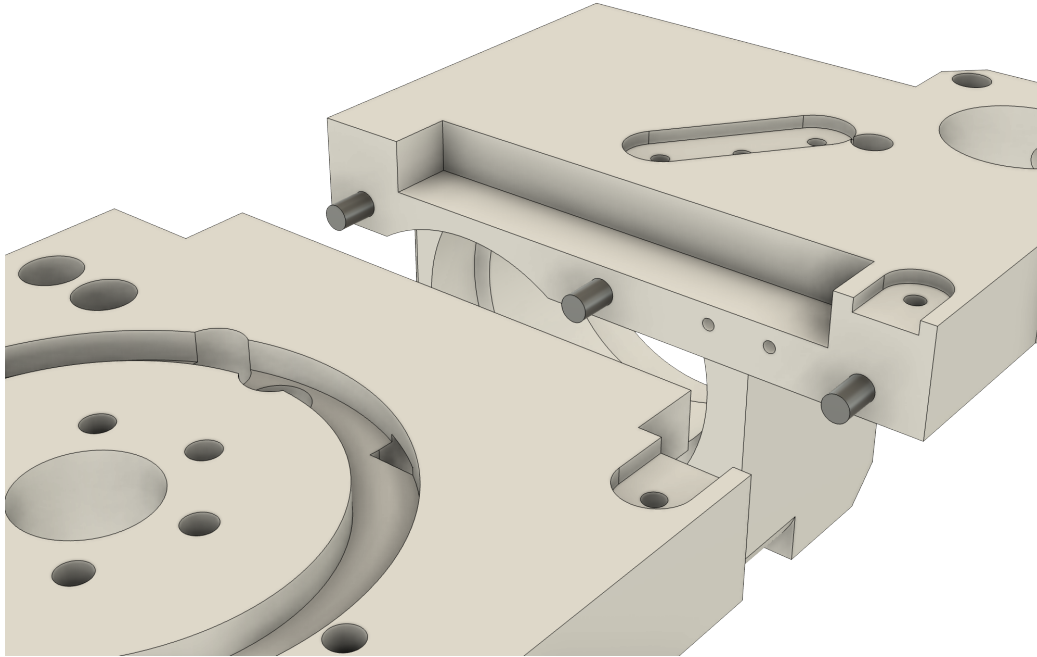
Ream all six 4 mm holes, three in each half.

Same technique as the arm: square to the face, steady hand pressure, clear chips often, clockwise in and clockwise out.

Do not confuse the reamers. The 4 mm one is for this part, the 6 mm one for the arm.

Test each hole with a 4 x 18 pin before moving on.

Step 2 - J1 turret - sand and degrease



This step

120 grit sandpaper
99% isopropyl alcohol
Nitrile gloves

Butt face, lap face,
tongue end and sides

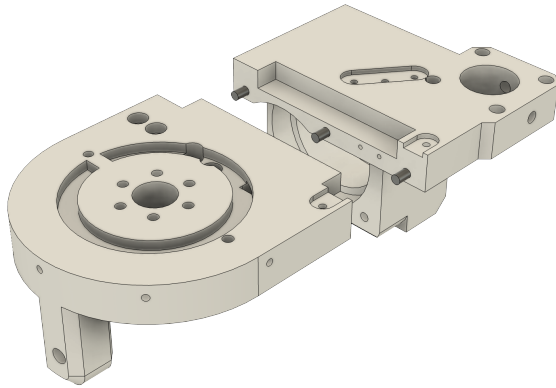
This part has four surfaces to prepare, not one: the lower butt face, the horizontal lap face, the tongue end face, and the tongue side walls.

Scuff every one of them with 120 grit until uniformly dull, then wipe with isopropyl and let it flash off.

Keep the paper out of the reamed pin holes.

Gloves on once the faces are clean.

Step 3 - J1 turret - dry fit



The notch is much harder to reach than the tongue. Small passes, re-test between each.

This step

J1_turret_rework_blocker_A_turret_end
J1_turret_rework_blocker_B_arm_end
3 x dowel pin 4 x 18 mm

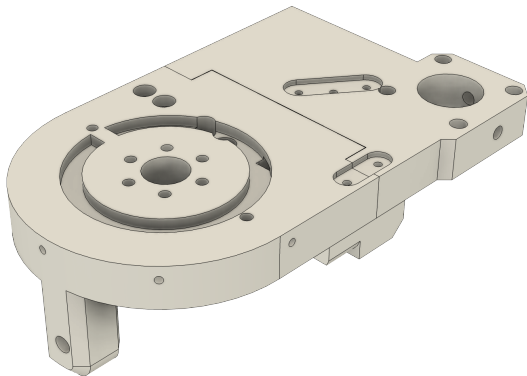
No glue at this stage

Seat all three pins in part B, which takes 12 mm of an 18 mm pin. The remaining 6 mm enters a 7 mm hole in A.

Bring the halves together. The lap tongue on A has to slide into the notch in B. It is modelled with 0.15 mm clearance on all three lap faces.

If it binds, sand the TONGUE, not the notch.

Step 4 - J1 turret - glue and clamp



Wipe squeeze out with isopropyl while it is still wet.

This step

Two part epoxy
3 x dowel pin 4 x 18 mm
Spring clamps

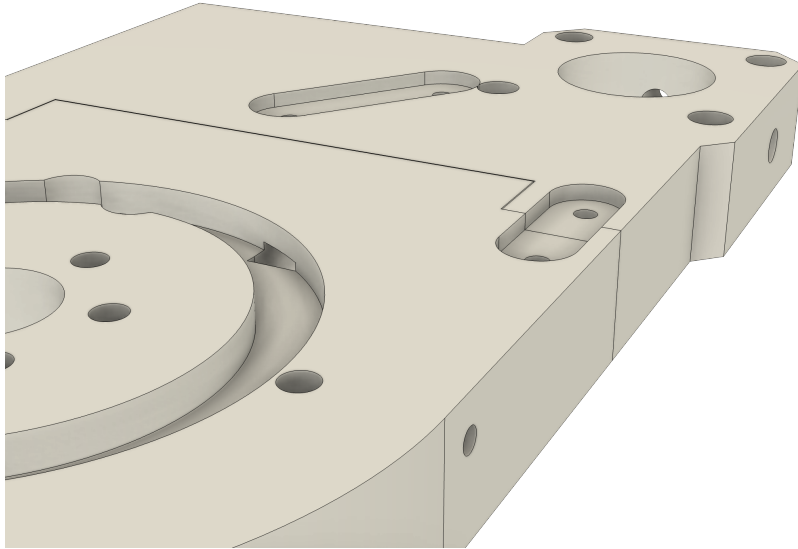
Set 4-6 h, full cure 24 h

Coat the butt face, the lap face and the tongue end. All three are structural.

Seat the pins in B, wipe a film on each, and do not fill the holes.

Close the joint and clamp along the part axis, straight across the butt seam. The lap step handles alignment in Z by itself, which is why this joint is more forgiving than the arm.

Step 5 - J1 turret - drill the pilot holes



Also check the large bore the seam crosses. Scrape any step or squeeze out flush.

This step

1.8 mm drill bit

Two pilots cross the seam

Drill AFTER full cure

Two 1.8 mm screw pilots run along the seam axis and are crossed by the cut. Their bores are geometrically unchanged, but drill them through only after the epoxy has fully cured.

Your printer cannot hold 1.8 mm anyway, so these needed drilling regardless. Doing it after the glue up makes the seam irrelevant to the result.

Use the printed holes as your guide and go slowly.